Exercises for *Secular Coefficients* mini-course

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1 Background

1.1 Number theory

1. Prove that $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ holds for $\Re s > 1$.
   Given a nontrivial Dirichlet character χ modulo m, prove that $L(s,\chi) = \prod_p (1-\chi(p)p^{-s})^{-1}$ for $\Re s > 1$.

2. Show that $\zeta(s) - 1/(s-1)$ has analytic continuation to $\Re s > 0$.
   Hint: express the difference as $1 - s\int_1^\infty \{t\} t^{-s-1} dt$ when $\Re s > 1$, where $\{t\} := t - \lfloor t \rfloor$ is the fractional part of t.

3. Given a nontrivial Dirichlet character χ modulo m, prove that $\sum_{a \in (\mathbb{Z}/m\mathbb{Z})^\times} \chi(a) = 0$.
   Prove that $\sum_{\chi \bmod m} \chi(a)\overline{\chi}(b) = \phi(m)\mathbf{1}_{a \equiv b \bmod m}$ where the sum is over all Dirichlet characters modulo m (here $(a,m) = (b,m) = 1$).

4. Given a nontrivial Dirichlet character χ modulo m, prove that $\sum_{n=1}^\infty \chi(n) n^{-s}$ converges (conditionally) for $\Re s > 0$.
   Hint: observe there is a constant C such that $|\sum_{n \le x} \chi(n)| \le C$ holds for all $x \ge 1$.

5. Let M be a polynomial in $\mathbb{F}_q[T] \setminus \mathbb{F}_q$. Given a nontrivial Dirichlet character χ modulo M, prove that $\sum_{A \in (\mathbb{F}_q[T]/M\mathbb{F}_q[T])^\times} \chi(A) = 0$.
   Deduce that $\mathcal{L}(u,\chi) = \sum_{f \text{ monic}} \chi(f) u^{\deg f}$ is a polynomial of degree at most $\deg M - 1$.
   Show that if $\mathcal{L}(u,\chi) = \prod_{i=1}^d (1-\gamma_i u)$ for complex numbers $\gamma_1, \ldots, \gamma_d$ then

$$\sum_{f \in \mathbb{F}_q[T],\, f \text{ monic}, \deg f = n} \chi(f)\Lambda(f) = -\sum_{i=1}^d \gamma_i^n$$

   where $\Lambda(f) = \deg P$ if f is a power of an irreducible polynomial P, and 0 otherwise.

1.2 Probability

1. If $X \sim N(0,1)$ (X is distributed like a standard real Gaussian) then, for every real a,

$$\mathbb{E}[e^{aX}] = e^{a^2/2}.$$

2. If $X \sim N_{\mathbb{C}}(0,1)$ (X is distributed like a standard complex Gaussian) then, for every real a, b,

$$\mathbb{E}[e^{aX + b\overline{X}}] = e^{ab}.$$

3. If $X \sim N_{\mathbb{C}}(0,1)$ then $\mathbb{E}[X^{k_1}\overline{X}^{k_2}] = (k_1)!$ if $k_1 = k_2$, and $\mathbb{E}[X^{k_1}\overline{X}^{k_2}] = 0$ otherwise.

2 Tutorial exercises

1. Prove that $\mathbb{E}[\alpha(n)\overline{\alpha}(m)] = 0$ if n, m are distinct positive integers, and $\mathbb{E}[\alpha(n)\overline{\alpha}(m)] = 1$ if $n = m$. Here α is the Steinhaus random multiplicative function.

2. Prove that $\mathbb{E}\left|\frac{1}{\sqrt{x}}\sum_{n\le x}\alpha(n)\right|^4 \sim C\log x$ for an explicit constant $C > 0$.
   Hint: parametrize solutions to $ab = cd$ by $a = ef$, $b = gh$, $c = eg$, $d = fh$ with $\gcd(f, g) = 1$.

3. Let $U_N \in U(N)$ be a random Haar-distributed unitary matrix. Define $\mathrm{Sc}_i(U_N)$ via $\det(I - uU_N) = \sum_{i=0}^{N}(-u)^i \mathrm{Sc}_i(U_N)$. For each $0 \le n \le N$ show that $|\mathrm{Sc}_n(U_N)|$ and $|\mathrm{Sc}_{N-n}(U_N)|$ have the same law.
   Hint: Use the functional equation.

4. Let $(N_i)_{i\ge1}$ be iid standard complex Gaussians. Let $F(u) = \exp(-\sum_{i\ge1} N_i u^i/\sqrt{i})$. Denote by c_n the coefficient of u^n in F. Prove that $\mathbb{E}|c_n|^4 = n + 1$.
   Hint: Consider $\mathbb{E}[F(u_1)F(u_2)\overline{F}(u_3)\overline{F}(u_4)]$. Can you compute higher moments? What do you get?

5. Compute $\mathbb{E}|L(1/2, \chi_M)|^2$ asymptotically (as $\deg M \to \infty$), where χ_M is a Dirichlet character chosen uniformly at random from the set of nontrivial Dirichlet characters modulo an irreducible $M \in \mathbb{F}_q[T]$. Can you compute $\mathbb{E}|\det(I - U_N)|^2$ where $U_N \in U(N)$ is a Haar-distributed unitary matrix?

6. Let χ_M be as in the last exercise. Recall that $\mathcal{L}(u, \chi_M) = \det(I - u\sqrt{q}\Theta_{\chi_M})$ for some matrix $\Theta_{\chi_M} \in U(\deg M - 1)$. Fix $k \ge 1$. Show that if $nk < \deg M$ then $\mathbb{E}\left|\frac{\mathrm{Tr}(\Theta_{\chi_M}^n)}{\sqrt{n}}\right|^{2k} \sim \mathbb{E}|N_{\mathbb{C}}(0,1)|^{2k}$ as $\deg M \to \infty$.

3 More

1. For x, y on the unit circle, estimate $\mathbb{E}[G_N(x)G_N(y)]$ where $G_N(x) = \Re\sum_{i=1}^{N}\frac{N_i}{\sqrt{i}}x^i$, and $(N_i)_i$ are iid standard complex Gaussians. For x, y with real part equal to $1/2$ estimate $\mathbb{E}[\Re(\log F_N(x))(\Re(\log F_N(y)))]$ where $F_N(s) = \prod_{p\le N}(1 - \alpha(p)p^{-s})^{-1}$, α the Steinhaus function.

2. (Bohr–Jessen in $\Re s > 1$.) Let α be the Steinhaus function and let $F(s) = \sum_{n\ge1}\alpha(n)/n^s$. Prove that $\zeta(s + iU_T) \xrightarrow{d} F(s)$ as $T \to \infty$, where s is fixed with $\Re s > 1$, and $U_T \sim \mathrm{Uniform}[0, T]$.

3. (Bohr–Jessen in $\Re s > 1$ for Dirichlet characters over a finite field.) Let χ_M be a Dirichlet character chosen uniformly at random from the set of nontrivial Dirichlet characters modulo an irreducible $M \in \mathbb{F}_q[T]$. As $\deg M \to \infty$, show that $L(u, \chi_M) \xrightarrow{d} L(u, \alpha_q)$ for every $|u| < 1/q$, where α_q is the Steinhaus random multiplicative function defined on monic polynomials.

4. The following is the most precise version of McLeish CLT.

   Theorem (McLeish). *Let $Z_1, \ldots, Z_N$ be a martingale difference sequence of complex-valued random variables, i.e. $S_N = \sum_{n=1}^{N} Z_n$ is a martingale. For any real $t_1, t_2 \in \mathbb{R}$ we have, with $t^2 = (t_1^2 + t_2^2)/2$,*

   $$\mathbb{E}[e^{it_1 Re(S_N) + it_2 Im(S_N)}] = e^{-t^2/2} + O\Big(e^{t^2}\Big(\sum_{n=1}^{N}\mathbb{E}[|Z_n|^4]\Big)^{\frac14}\Big) + O\Big(e^{t^2}\Big(\mathbb{E}\Big[\Big(\sum_{n=1}^{N}|Z_n|^2 - 1\Big)^2\Big]\Big)^{\frac12}\Big)$$
   $$+ O\Big(e^{t^2}\max_{\phi}\Big(\mathbb{E}\Big[\Big(\sum_{n=1}^{N}(e^{-i\phi}Z_n^2 + e^{i\phi}\overline{Z_n}^2)\Big)^2\Big]\Big)^{\frac12}\Big).$$

   Deduce the criterion of Soundararajan–Xu: let $\mathcal{A}_x \subseteq \{1, 2, \ldots, x\}$ and set $S(x, \mathcal{A}_x) = \sum_{n\in\mathcal{A}_x}\alpha(n)/\sqrt{|\mathcal{A}_x|}$. Then $S(x, \mathcal{A}_x)$ tends to a standard complex Gaussian if two conditions are met: $\lim_{x\to\infty}\mathbb{E}|S(x, \mathcal{A}_x)|^4 = 2$, and $|\{n \in \mathcal{A}_x : P(n) = p\}| = o(|\mathcal{A}_x|)$ for every $p \le x$ ($P(n)$ is the largest prime factor of n).

5. Let $b(\lambda)$ be a function on partitions. Given a partition λ, we denote by $a_i(\lambda)$ the number of parts of λ of size i. Let $S_N = \sum_{\lambda\vdash N} b(\lambda)\prod_{i=1}^{N} N_i^{a_i(\lambda)}$. Using McLeish CLT, establish a sufficient criterion for S_N to tend in distribution to a standard complex Gaussian.